

## EDUCATION

- 2026.03–Present**   **Ph.D. Student in Data Science**  
*Graduate School of Data Science, Kyungpook National University, Korea*  
Advisor: Prof. Suhyeon Kim
- 2024.03–2026.02**   **M.S. in Data Science**  
*Graduate School of Data Science, Kyungpook National University, Korea*  
Dissertation: *A Newtonian-informed Transformer AutoEncoder for Vessel Navigation Risk Index Estimation*  
Advisor: Prof. Suhyeon Kim
- 2017.03–2024.02**   **B.S. in Mathematics**  
*Pusan National University, Korea*
- 2022.03–2024.02**   **Interdisciplinary Major in Industrial Mathematics & Software**  
*Pusan National University, Korea*
- 2023.03–2023.12**   **Undergraduate Researcher**  
*Industrial Mathematics Center, Pusan National University, Korea*

## RESEARCH INTERESTS

- AI for mobility, industrial systems, and medical data
- Federated learning, privacy-preserving AI, and cybersecurity
- Risk estimation, anomaly detection, and decision-support modeling
- Graph machine learning and spatiotemporal representation learning

## PUBLICATIONS

**Notation.** <sup>†</sup> equal contribution; \* corresponding author.

### SCI(E) / SSCI Journal Papers

- [1] Eun-ji Lee<sup>†</sup>, **Byeongkeun Kwon**<sup>†</sup>, Suhyeon Kim\*, Sung Hyun Noh\*, “Predictive Modelling of the Need for Tracheostomy after Traumatic Cervical Spinal Cord Injury using Machine Learning,” *Yonsei Medical Journal*, 2026. Co-first author; Accepted; JCR Q1.
- [2] **Byeongkeun Kwon**<sup>†</sup>, Dain Lee<sup>†</sup>, Hyunjun Ko, Hanbin Lee, Hyeonjun Hwang\*, Suhyeon Kim\*, “A Graph Neural Network Approach for Aquatic Biodiversity Prediction Leveraging Water System Interconnections,” *Ecological Informatics*, May 2026. Co-first author; JCR Q1, 4.1%.

### Scopus Papers

- [1] **Byeongkeun Kwon**, Suhyeon Kim\*, “Driver Behavior Anomaly Detection based on Federated Learning Considering Data Distribution Imbalance,” *The Journal of Korean Institute of Communications and Information Sciences*, 50(3), 395–405, March 2025. First author; Best Paper of JKICS.

### Working Papers

- [1] **Byeongkeun Kwon**, Dain Lee, Suhyeon Kim\*, “A Newtonian-informed Transformer Autoencoder for Fishing Vessel Navigation Risk Index Estimation.” Under review at *Ocean Engineering*.
- [2] **Byeongkeun Kwon**<sup>†</sup>, Dasom Seong<sup>†</sup>, Jiyun Park, Hyeonjun Hwang\*, Suhyeon Kim\*, “Water Quality Risk Index Estimation using Graph Neural Networks for Spatiotemporal Watershed Systems.” Submitted to *Water Research*.

- [3] **Byeongkeun Kwon**<sup>†</sup>, Dasom Seong<sup>†</sup>, Sung Hyun Noh<sup>\*</sup>, Suhyeon Kim<sup>\*</sup>, “SSIU-Net: Spine Screw Inverse U-Net for Anatomical Representation-guided Pedicle Screw Trajectory Planning.” Manuscript in preparation. Target journal: *Medical Image Analysis*.
- [4] **Byeongkeun Kwon**<sup>†</sup>, Sesun You<sup>†</sup>, Young Seop Son<sup>\*</sup>, Suhyeon Kim<sup>\*</sup>, “CAN-based Heavy-Duty Truck Weight Estimation: A Dual-phase Analysis of AI and Vehicle Dynamics.” Manuscript in preparation. Target journal: *IEEE Transactions on Intelligent Vehicles*.

## CONFERENCE PRESENTATIONS

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### International Conferences

1. **Byeongkeun Kwon**, Dasom Seong, Suhyeon Kim, Sung Hyun Noh, “3D Generative Models for Vertebral Segmentation and Pedicle Screw Path Prediction,” *MICCAI 2025 CLINICCAI*, Daejeon, Korea, September 23–27, 2025.
2. **Byeongkeun Kwon**, Eun-ji Lee, Suhyeon Kim, Sung Hyun Noh, “Development of Machine Learning Model Predicting the Need for Tracheostomy after Cervical Spinal Cord Injury,” *ASIA SPINE & NCS 2025*, Busan, Korea, September 4–6, 2025.
3. Hyosung Kim, Taejun Kwon, **Byeongkeun Kwon**, Jieun Jang, Jihun Jang, “DRIVE: Drowsiness Recognition and Intelligent Vibration Evocation,” *IEEE World Haptics Conference 2025*, Suwon, Korea, July 8–11, 2025.
4. **Byeongkeun Kwon**, Suhyeon Kim, “Classical Mechanics Meet Deep Learning: Ship Hazardous Navigation Detection System,” *IISE Annual Conference & Expo 2025*, Atlanta, USA, May 31–June 3, 2025. *Presenter; Session Chair*.
5. **Byeongkeun Kwon**, Suhyeon Kim, “Baseball Match Prediction based on Graph Neural Network,” *Deep Learn 2024: 11th International School on Deep Learning*, Porto, Portugal, July 15–19, 2024.

### Domestic Conferences

1. **Byeongkeun Kwon**, Dasom Seong, Suhyeon Kim, “Graph Neural Network-based Water Quality Risk Index for Spatio-Temporal Watershed Systems,” *Proceedings of KIIE 2026 Joint Conference*, Gyeongju, Korea, June 4–6, 2026.
2. **Byeongkeun Kwon**, Seseun You, Dain Lee, Dasom Seong, Suhyeon Kim, Hyung Jong Cho, Hyuntae Cho, Young Seop Son, Suhyeon Kim, “Lightweight MLP-Mixer for Real-time Weight-in-Motion with Embedded Deployment,” *Proceedings of KSAE 2026 Spring Conference*, Jeju, Korea, May 20–23, 2026.
3. **Byeongkeun Kwon**, Dasom Seong, Suhyeon Kim, Sung Hyun Noh, “3D Generative Models for Vertebral Segmentation and Pedicle Screw Path Prediction,” *The 44th Annual Spring Meeting of the Korean Neurosurgical Society*, Jeju, Korea, April 16–18, 2026.
4. **Byeongkeun Kwon**, Dain Lee, Dasom Seong, Suhyeon Kim, “Development of a Generative Adversarial Networks-based Fuzzing Test Technique for Vehicle Communication Security,” *Proceedings of KSAE 2025 Fall Conference*, Busan, Korea, November 12–15, 2025.
5. **Byeongkeun Kwon**, Suhyeon Kim, “A Newtonian-informed Transformer AutoEncoder for Vessel Navigation Risk Estimation,” *Proceedings of KIIE 2025 Fall Conference*, Daejeon, Korea, November 6–7, 2025.
6. **Byeongkeun Kwon**, Suhyeon Kim, “Mixer-integrated Graph Neural Networks for Accurate and Efficient Prediction of Crystalline Solid Properties,” *Proceedings of KIIE 2025 Joint Conference*, Jeju, Korea, June 18–21, 2025. *Poster*.
7. **Byeongkeun Kwon**, Suhyeon Kim, “Privacy-preserving Federated Learning for Abnormal Driving Behavior Detection,” *Proceedings of KICS Fall Conference 2024*, Gyeongju, Korea, November 20–22, 2024.
8. Eun-ji Lee, **Byeongkeun Kwon**, Suhyeon Kim, Sung Hyun Noh, “Development of Machine Learning Model Predicting the Need for Tracheostomy after Cervical Spinal Cord Injury,” *NCS 2024*, Seoul, Korea, September 5–7, 2024.
9. **Byeongkeun Kwon**, Suhyeon Kim, “Baseball Match Prediction based on Graph Neural Network,” *Proceedings of Korean Artificial Intelligence Association 2024 Conference*, Busan, Korea, August 16–17, 2024. *Poster*.

# AWARDS AND HONORS

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## Grants

**2026 Spring**      **KNU Ph.D. Fellow**, Kyungpook National University. Award amount: KRW 5,000,000 (approximately USD 3.3K).

## Awards

- 2025**      **Minister of Science and ICT Award**, 2025 K-Data Science Conference & Hackathon, Hackathon Track.
- 2025**      **Minister of Science and ICT Award**, 2025 ICT Challenge.
- 2025**      **Minister of Environment Award**, 2025 Environmental Data Utilization & Analysis Competition, Analysis Track.
- 2024**      **President Award of NIA**, 2024 K-Data Science Conference & Hackathon, Hackathon Track.
- 2024**      **Chairperson Award of Korea National Park Service**, 2024 Environmental Data Utilization & Analysis Competition, Analysis Track.
- 2024**      **Finalist Award**, 2024 Weather Big Data Contest, Fog Track.
- 2024**      **Finalist Award**, 2024 Weather Big Data Contest, Power Track.
- 2023**      **Finalist Award**, 2023 Weather Big Data Contest.
- 2023**      **Encouragement Award**, 2023 Yeongnam Mathematical Society, Poster Presentation.
- 2023**      **Excellence Award & Encouragement Award**, 2023 Mathematics May Competition, Departmental Contest.

# PATENTS

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- 1. Aquatic biodiversity prediction system and its method, 2025.

# TEACHING ASSISTANTSHIPS

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- 2026 Spring**      **Teaching Assistant**  
*Data Science Deep Learning* (DASC0735), Graduate School of Data Science, Kyungpook National University
- 2026 Spring**      **Teaching Assistant**  
*Graph Data Analysis* (DASC0767), Graduate School of Data Science, Kyungpook National University

# ACADEMIC SERVICE

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- 2025**      Session Chair, *IISE Annual Conference & Expo 2025*, Atlanta, USA.

sectionResearch Experience and Projects

## Ongoing Projects

- 2026.03–2031.02**      **Development of Graph Deep Learning Core Technologies for Hypertrustworthy Dynamic Structural AI Countering Cybersecurity Threats**  
National Research Foundation of Korea, Individual Basic Research / Excellent Young Researcher Type B Researcher

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|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>2026.04–2028.03</b> | <b>Advancement of an Adaptive AI-Based Industrial Control Protocol Security Testing Automation Platform and Expansion into Industrial Domains</b><br>Industry-Academia-Research Collabo R&D Program, Commercialization R&D Program | Researcher |
| <b>2025.03–2027.08</b> | <b>BK21 FOUR Project: GLOW-AI Innovative Talent Development Education and Research Group</b><br>BK21 FOUR                                                                                                                          | Researcher |
| <b>2023.06–2026.05</b> | <b>Sequential Data Analysis using Multi-view Graph Deep Representation Learning and Its Industrial Applications</b><br>National Research Foundation of Korea, Creative Challenge Research Support Program                          | Researcher |
| <b>2024.07–Present</b> | <b>Data-driven Intelligent Mobility Research Center</b><br>IITP, ICT Research Center / ITRC Program                                                                                                                                | Researcher |
| <b>2023.09–Present</b> | <b>Regional Demand Oriented Data Science Human Resource Development for Regional Industry Innovation</b><br>Data Science Convergence Human Resource Development Program                                                            | Researcher |

### Completed Projects

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|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>2025.05–2025.12</b> | <b>Generative AI Platform for Automated Analysis of Industrial Control Protocol Specifications and Automated Security Vulnerability Test-case Generation</b><br>Industry-Academia-Research Collabo R&D Program, Preliminary Research Program  | Researcher |
| <b>2024.08–</b>        | <b>GAN-based Automated Test-case Generation and Security Vulnerability Detection for Vehicle Communication Protocols and Infotainment File Formats</b><br>Korean Ministry of SMEs and Startups, Startup Growth Technology Development Program | Researcher |

### REFERENCES

**Prof. Dr. Suhyeon Kim** | Kyungpook National University  
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 Graduate School of Data Science  
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 Webpage: <https://sites.google.com/view/suhyeon-kim/home>